CSE 3204
Formal Language, Automata and Computability

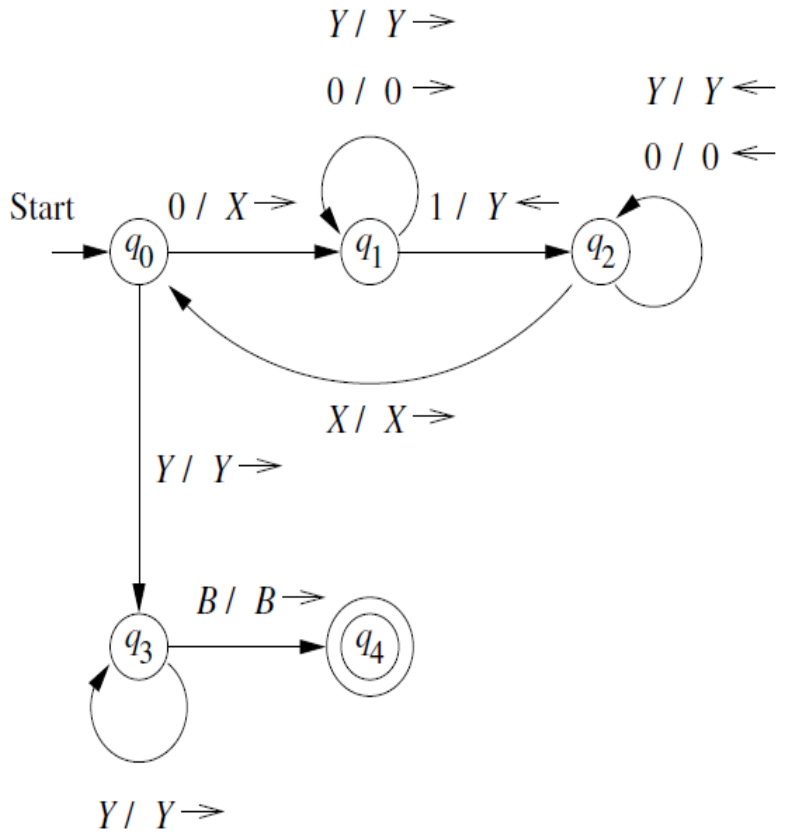
Lecture 20
Turing Machine

Transition Diagram of a TM

$$M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0,1\}, \{0,1,X,Y,B\}, \delta, q_0, B, \{q_4\})$$

State	Symbol 0	1	X	Y	B
q_0	(q_1, X, R)	–	–	(q_3, Y, R)	–
q_1	$(q_1, 0, R)$	(q_2, Y, L)	–	(q_1, Y, R)	–
q_2	$(q_2, 0, L)$	–	(q_0, X, R)	(q_2, Y, L)	–
q_3	–	–	–	(q_3, Y, R)	(q_4, B, R)
q_4	–	–	–	–	–

Figure 8.9: A Turing machine to accept $\{0^n1^n \mid n \geq 1\}$



Transition diagram for a TM that accepts strings of the form 0^n1^n

Turing Machine Computing $m \dot{-} n = \max(m - n, 0)$

monus or *proper subtraction* and is defined by $m \dot{-} n = \max(m - n, 0)$. That is, $m \dot{-} n$ is $m - n$ if $m \geq n$ and 0 if $m < n$.

A TM that performs this operation is specified by

$$M = (\{q_0, q_1, \ldots, q_6\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B)$$

tape consisting of $0^m 1 0^n$ surrounded by blanks. M halts with $0^{m \dot{-} n}$ on its tape, surrounded by blanks.

Moves of the TM

M repeatedly finds its leftmost remaining 0 and replaces it by a blank. It then searches right, looking for a 1. After finding a 1, it continues right, until it comes to a 0, which it replaces by a 1. M then returns left, seeking the leftmost 0, which it identifies when it first meets a blank and then moves one cell to the right. The repetition ends if either:

1. Searching right for a 0, M encounters a blank. Then the n 0's in 0^m10^n have all been changed to 1's, and $n+1$ of the m 0's have been changed to B. M replaces the $n+1$ 1's by one 0 and n B's, leaving $m-n$ 0's on the tape. Since $m \geq n$ in this case, $m - n = m \dot{-} n$.

2. Beginning the cycle, M cannot find a 0 to change to a blank, because the first m 0's already have been changed to B. Then $n \geq m$, so $m \dot{-} n = 0$. M replaces all remaining 1's and 0's by B and ends with a completely blank tape.

Transition Table of TM

State	Symbol 0	1	B
q_0	(q_1, B, R)	(q_5, B, R)	–
q_1	$(q_1, 0, R)$	$(q_2, 1, R)$	–
q_2	$(q_3, 1, L)$	$(q_2, 1, R)$	(q_4, B, L)
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
q_4	$(q_4, 0, L)$	(q_4, B, L)	$(q_6, 0, R)$
q_5	(q_5, B, R)	(q_5, B, R)	(q_6, B, R)
q_6	–	–	–

Recursively Enumerable Language or RE Language

More formally, let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ be a Turing machine. Then $L(M)$ is the set of strings w in Σ^* such that $q_0 w \overset{*}{\vdash} \alpha p \beta$ for some state p in F and any tape strings α and β. This definition was assumed when we discussed

Turing Machine and Halting

machines: acceptance by halting. We say a TM *halts* if it enters a state q, scanning a tape symbol X, and there is no move in this situation; i.e., $\delta(q, X)$ is undefined.

We can always assume that a TM halts if it accepts. That is, without changing the language accepted, we can make $\delta(q, X)$ undefined whenever q is an accepting state. In general, without otherwise stating so:

- We assume that a TM always halts when it is in an accepting state.

Notational Conventions for Turing Machines

The symbols we normally use for Turing machines resemble those for the other kinds of automata we have seen.

1. Lower-case letters at the beginning of the alphabet stand for input symbols.
2. Capital letters, typically near the end of the alphabet, are used for tape symbols that may or may not be input symbols. However, B is generally used for the blank symbol.
3. Lower-case letters near the end of the alphabet are strings of input symbols.
4. Greek letters are strings of tape symbols.
5. Letters such as q, p, and nearby letters are states.

Turing Machine

- A **mathematical model of computation**, representing the fundamental computational capabilities of a classical computer.
- An abstract machine used to study what can and cannot be computed.

Storage in a State

- The finite control can contain the state q and hold a finite amount of data.
- Do not need extension of the traditional TM.
- This representation allows to describe transitions in a more systematic way.

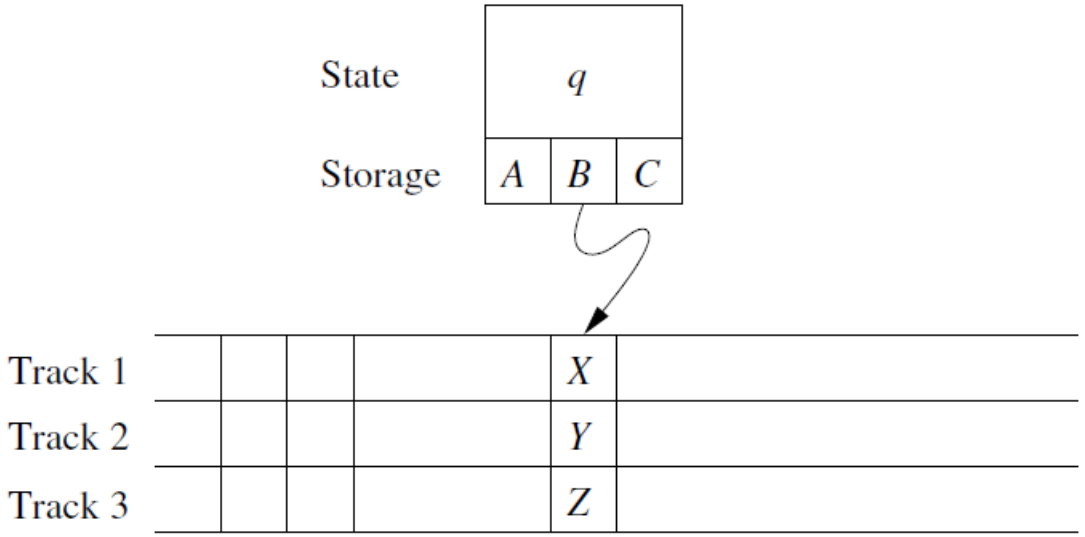


Figure 8.13: A Turing machine viewed as having finite-control storage and multiple tracks

TM Accepting Language $01^*+ 10^*$

$$M = (Q, \{0,1\}, \{0,1,B\}, \delta, [q_0, B], B, \{[q_1, B]\})$$

The set of states Q is $\{q_0, q_1\} \times \{0, 1, B\}$. That is, the states may be thought of as pairs with two components:

a) A control portion, q_0 or q_1, that remembers what the TM is doing. Control state q_0 indicates that M has not yet read its first symbol, while q_1 indicates that it *has* read the symbol, and is checking that it does not appear elsewhere, by moving right and hoping to reach a blank cell.

b) A data portion, which remembers the first symbol seen, which must be 0 or 1. The symbol B in this component means that no symbol has been read.

TM Accepting Language $01^*+ 10^*$

The transition function δ of M is as follows:

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$. Initially, q_0 is the control state, and the data portion of the state is B. The symbol scanned is copied into the second component of the state, and M moves right, entering control state q_1 as it does so.

2. $\delta([q_1, a], \overline{a}) = ([q_1, a], \overline{a}, R)$ where $\overline{a}$ is the "complement" of a, that is, 0 if $a = 1$ and 1 if $a = 0$. In state q_1, M skips over each symbol 0 or 1 that is different from the one it has stored in its state, and continues moving right.

3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$. If M reaches the first blank, it enters the accepting state $[q_1, B]$.

Multiple Tracks

Another useful "trick" is to think of the tape of a Turing machine as composed of several tracks. Each track can hold one symbol, and the tape alphabet of the TM consists of tuples, with one component for each "track." Thus, for instance,

Does not need the extension of the traditional TM.

Example 8.7 covered from the book.

- Section 8.2 and 8.3 of Chapter 8